



**Middlesex
University
London**

Superstore Analysis

By Amir Lorvand
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Introduction

The Sample – Superstore is transactional data that is commonly used in applying business intelligence as well as retail analysis. It depicts the sales activities of a made-up superstore and provides a sweeping, cross-sectional view of a host of variables ranging from customer profile attributes, to purchase specifics and product performance. It has 9,994 records and 21 attributes; therefore, it is ideal for the assessment of patterns important in strategic planning.

This dataset is obtained from Kaggle and Tableau's official website where such a data set is used as practice data for data visualization and analysis. This makes a structured design advantageous since a wide range of analysis methods can be employed on its numerous and diverse variety of variables. Attributes include categorical variables (e.g., Region, Category), numerical variables (e.g., Sales, Profit), and temporal variables (e.g., Order Date, Ship Date).

Before loading it in Tableau, the cleanliness of the accuracies, completeness, consistency, uniformity, and validity of the dataset was checked in Excel. In compliance with the analysis previously described, the dataset was identified as having no missing or invalid data outputs. However, potential cleaning steps that could be performed to ensure data integrity include:

Date Standardization: Specify the format of Order Date and Ship Date to a format convenient for timeserial analysis (as MM/DD/YYYY).

Normalization: Standardizing numerical columns such as Sales and Profit to avoid skewing results in algorithms or visualizations.

Duplicate Removal: Although duplicates are not present in this dataset, reviewing for repeated entries (e.g., identical Order ID) could ensure accurate aggregations.

Validation of Categorical Data: Checking for potential typos or inconsistencies in Region, Category, or Segment to ensure uniform grouping during analysis.

The variables of this dataset make it suitable for sales analysis, market segmentation, and sales profitability. If there are any potential cleaning requirements and by applying tools of the advanced visualization, the analysis can provide useful business intelligence information to help in the managerial decision-making process.

Data Analysis

The central region contributes to the sales but has a significantly lower profit ratio than the other regions (Figure 1). This section investigates key factors causing this issue, including discounts, product categories, customer segments, shipping methods, seasonal trends, and cost.

As can be seen in Figure 1 the central region contributes 21.82% of total sales, one of the key revenue drivers. However, its share of profit is only 13.86%, significantly lower than regions with similar lower sales contribution.

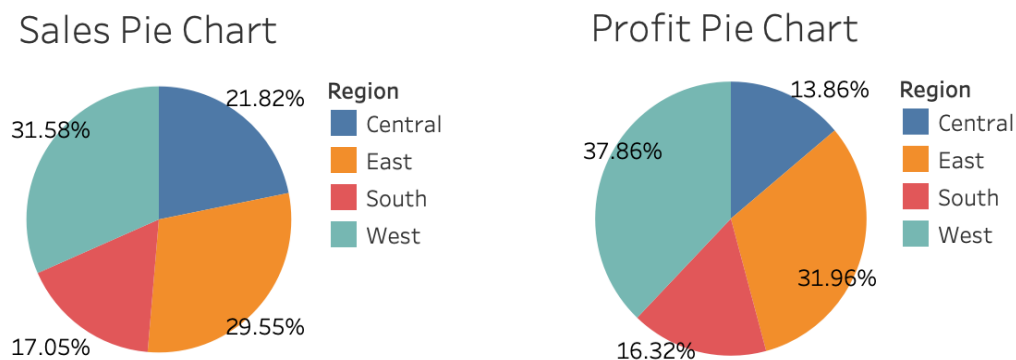


Figure 1: Sales and Profit Distribution

The profit ratio by region bar chart (Figure 2) shows that the central region has the lowest profit ratio despite contributing to a considerable share of total sales (as seen in Figure 1).

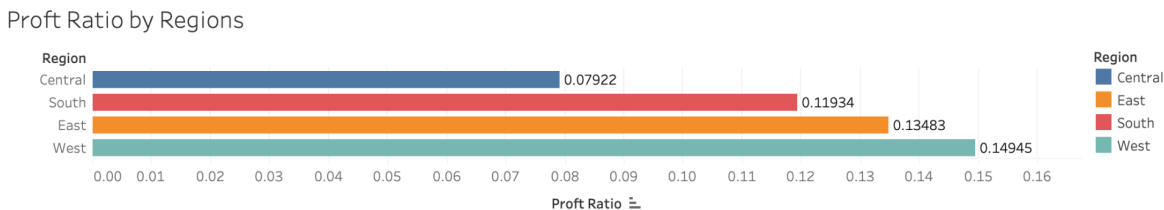


Figure 2 : Profit Ratio by Regions

This section investigates key factors causing this issue, including costs, customer segments, discounts, product categories, and seasonal trends.

Cost to Sale Ratio by Region

In the following bar chart (Figure 3) where the relationship between cost and sales is analysed in each region, the central region has the highest cost-to-sale ratio among all regions, approaching 0.95. This means that 95% of the sales revenue is being consumed by costs, leaving only a small margin for profit.

Cost-to-Sale Ratio

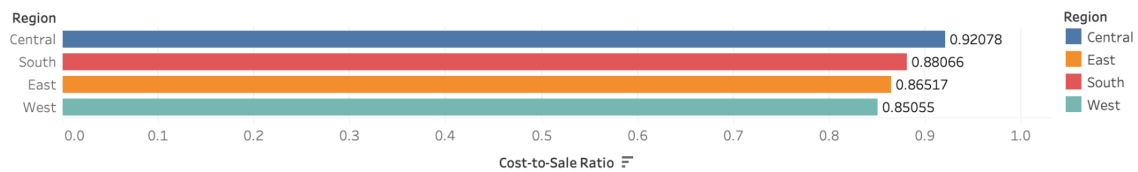


Figure 3 : Cost-to-Sale Ratio

The high cost-to-sales ratio directly contributes to the central region's low profit ratio. Compared to other regions, the central region operates with much tighter margins, making it more vulnerable to inefficiencies like aggressive discounting and shipping costs. It is recommended to adjust pricing strategies for high-cost items to ensure that sales cover costs with a reasonable margin along with introducing dynamic pricing models that consider costs and market demands.

Customer Segment Challenges

The following bar chart (Figure 4) of profit ratios by segment in the central region shows that corporate and home office segments are more profitable than consumer segments, while consumer segments generate the highest sales, indicating higher price sensitivity.

Customer Segments

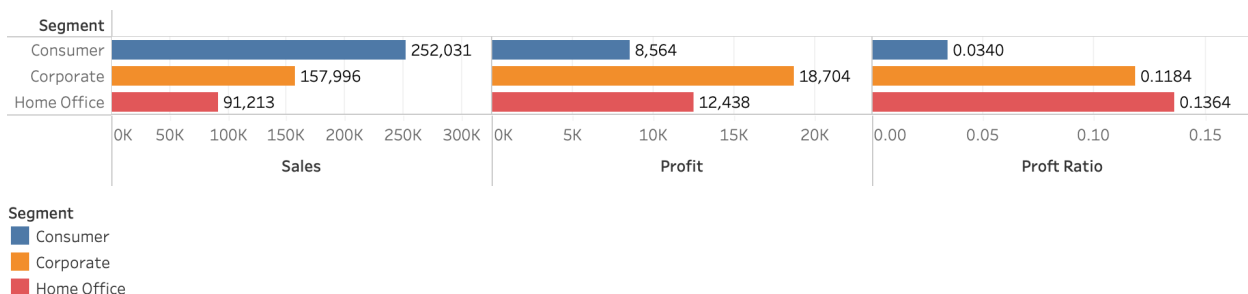


Figure 4 : Customer Segments Sales, Profit and Profit Ratio

To shift sales to more profitable customer segments, it is suggested that the company should target more corporate and home office markets because it offers higher profit margins. They should encouraged to purchase and use high products.

Discounts

Discounts in the central region are very high with particular transactions having discounts of up to 4 (Figure 5). Higher discounts are found to be directly linked with negative gross profit and thus have negative impacts on total profitability. As seen in Figure 4, where discount and profit were analysed, it can be discerned from the scatter plot that profits tend to go down with an increase in the discounts offered.

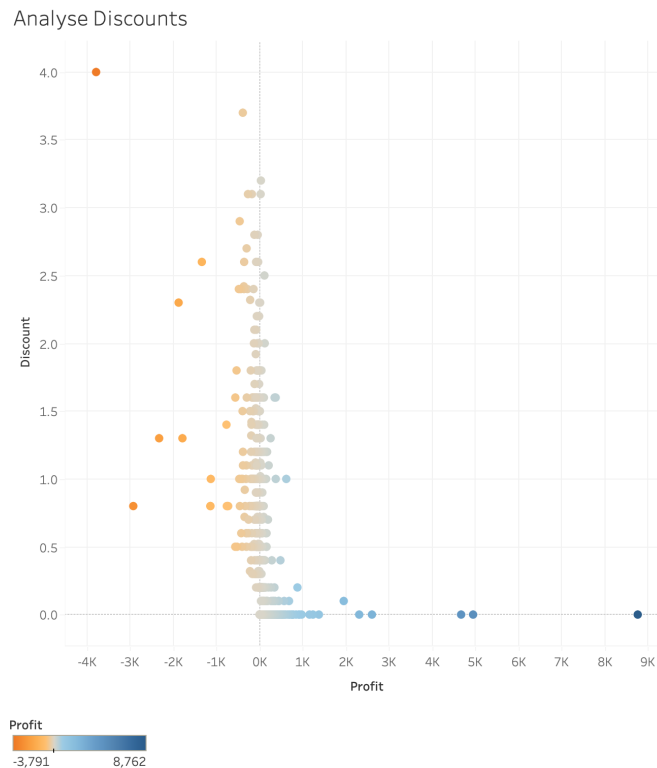


Figure 5 : Profit vs. Discounts

To optimise the discounting strategies, there should be a maximum of discounts, e.g. 25% while regularly monitoring the profitability of heavily discounted items. They help to minimise profit shrinkage from multiple sales promotions while lowering product prices.

Underperforming Categories

A bar chart of sales and profit by product sub-category in the central region (Figure 6) highlights all the sub-categories of furniture -except chairs- generate high sales but incur a negative profit margin. Also, office supplies have high sales but negative profits. Tables and other underperforming sub-categories drag down the region's overall profitability, despite their sales contribution.

Examine Product Category

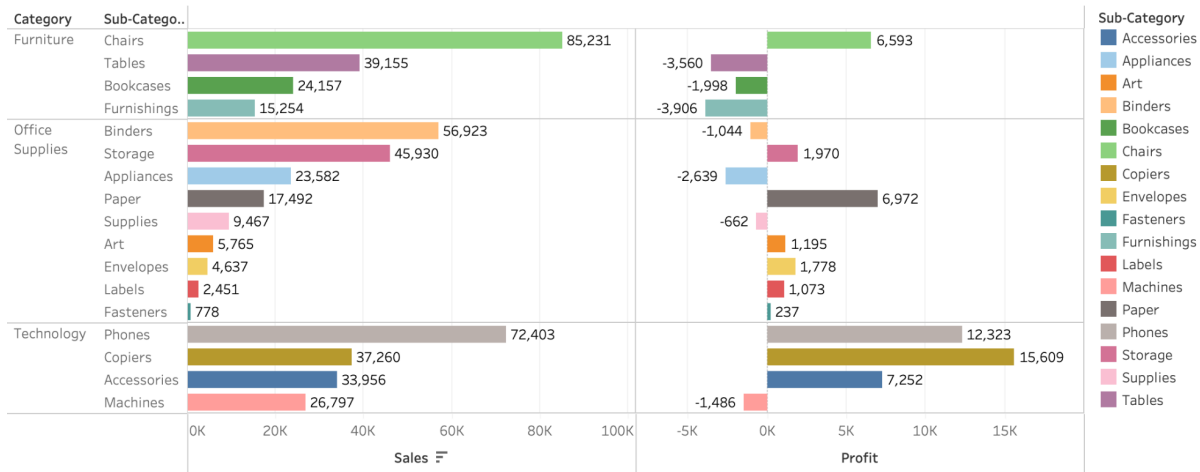


Figure 6 : performance of sub-categories in the central region

To improve profit margins in the furniture and office supplies category and boost overall profitability, the company can reevaluate pricing for underperforming sub-categories, or explore cost-reduction strategies for manufacturing or sourcing these products. Another way is to shift marketing toward high-margin products such as phones and copiers.

Seasonal Volatility

A line graph of the profit ratio trend over time (Figure 7) illustrates the profit ratio fluctuates significantly over time, with sharp declines observed in certain quarters. Seasonal inefficiencies or misaligned strategies may be contributing to profitability issues.

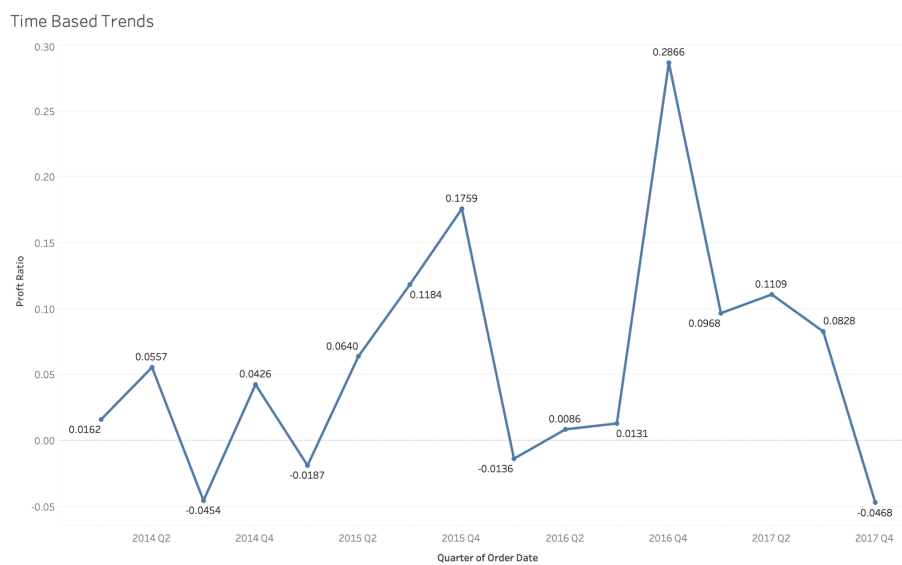


Figure 7 : Profit ratio over time

To smooth profit trends and reduce seasonal volatility, it is recommended to analyse operational strategies during low-performing quarters and align promotional campaigns with historically high-performance periods.

The analysis revealed that the central region’s low-profit ratio comes from a combination of underperforming categories, seasonal volatility, and especially discounting. However, this issue may not be limited to the central region. A detailed analysis of discounting across all regions is essential to pinpoint specific inefficiencies at the product level. The next step is to answer this question: How do discounts impact profitability at the sub-category level?

Average Discount vs. Profit

Figure 8 represents sub-categories like tables and supplies that result in negative profits of -17,725\$ and -1,189\$. But, binders, copiers, and phones maintain positive profits despite moderate discounts. It shows that excessive discounts severely impact profitability for some sub-categories.

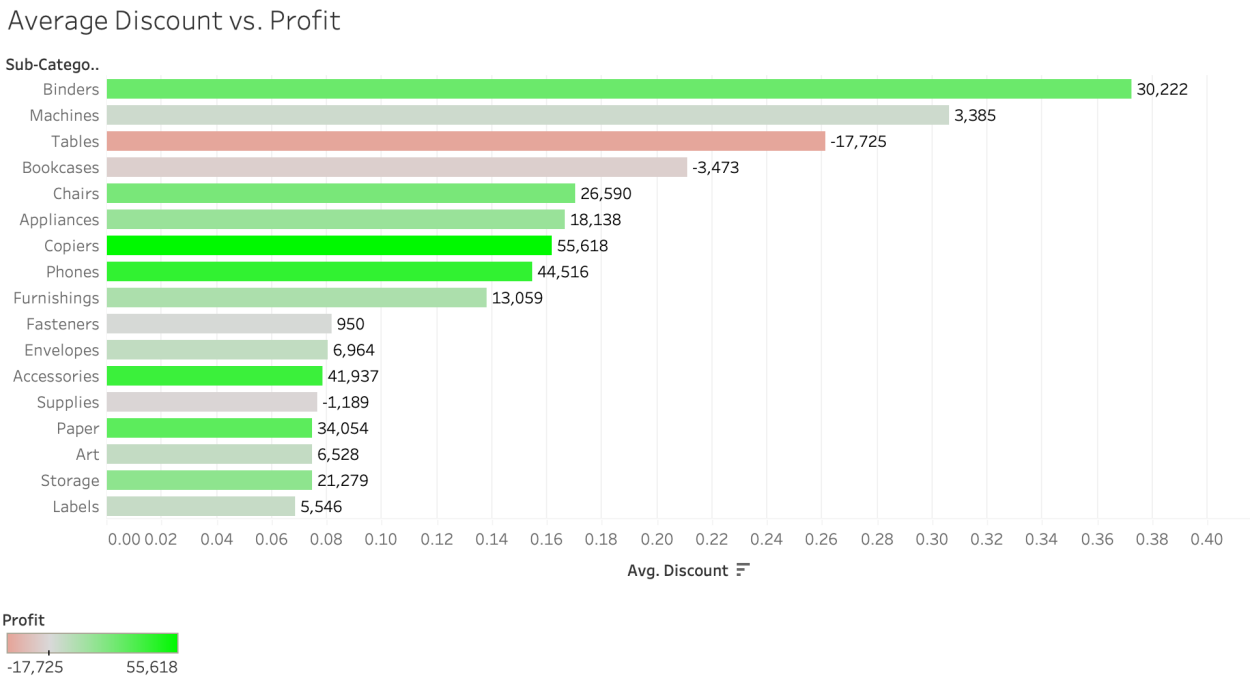
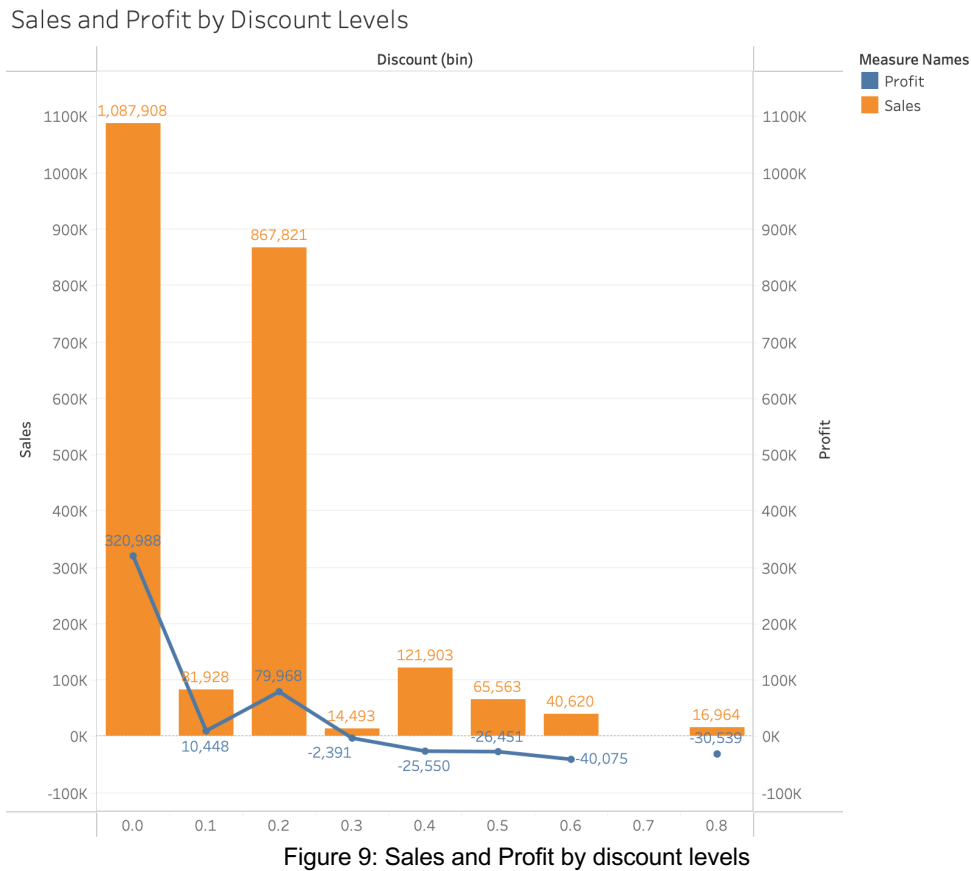


Figure 8: Average Discount vs. Profit (Ordered by Average Discount)

Reducing discount levels for underperforming sub-categories, like tables and supplies, and maintaining reasonable discounts for profitable ones, such as binders, would improve profitability.

Sales and Profit by Discount Levels

Figure 9 shows sales peak within 0 to 20% discount levels but decrease sharply beyond 30%. Profits also drop significantly, turning negative at discount levels above 40%. It represents that moderate discounts are effective for profits.



Capping discounts at 20% and reevaluating pricing for products with frequent high discounts would optimise both sales and profits.

Discount vs. Profit Ratio

Figure 10 highlights that sub-categories like tables and bookcases, with discounts above 20% have negative profit ratios. Meanwhile, categories such as copiers, phones, and accessories maintain positive profit ratios with moderate discounts. But, binders have a good profit ratio with a high discount level. It confirms that often high discounts harm underperforming sub-categories while others thrive with controlled discounts.

Discount vs. Profit

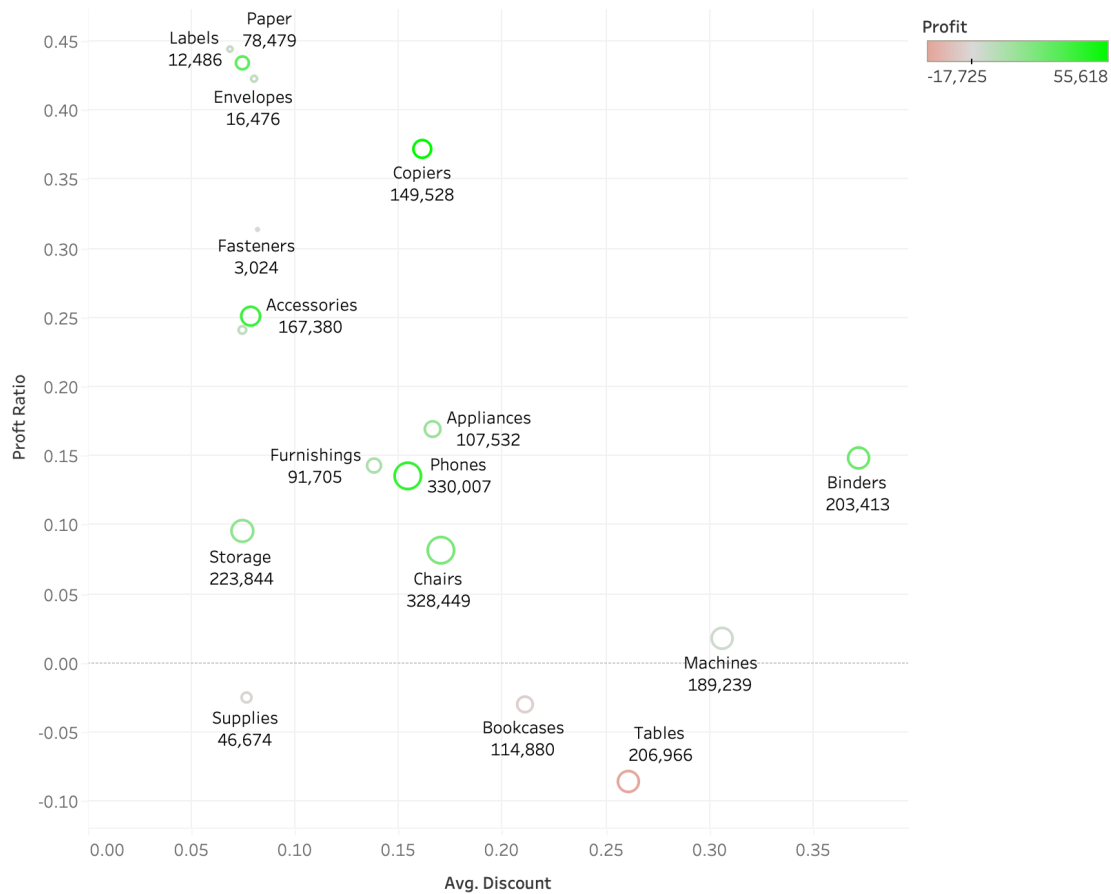


Figure 10: Discount vs. Profit Ratio

Limiting discounts for low-profit sub-categories and focusing promotions on profitable ones like copiers would enhance overall performance.

Discounts by Sub-Category and Region

The heat map (Figure 11) revealed regional disparities and large variations within sub-categories such as a much higher discount for tables and binders compared to other regions in the central and east regions. This means that over-discounting leads to profitability challenges in these regions. Exemplifying discount policies in all zones and increasing price cuts for comparatively poor sub-categories, such as tables in heavily discounted zones, would enhance profitability.

Discounts by Sub-Category and Region

Sub-Catego..	Region				Avg. Discount	
	Central	East	South	West		
Accessories	0.1246	0.0854	0.0859	0.0364	0.0190	0.5093
Appliances	0.4488	0.0730	0.1111	0.0309		
Art	0.1227	0.0693	0.1014	0.0312		
Binders	0.5093	0.3534	0.3756	0.2817		
Bookcases	0.2328	0.2200	0.1000	0.2288		
Chairs	0.1929	0.1506	0.0977	0.2000		
Copiers	0.1125	0.2500	0.0857	0.1440		
Envelopes	0.1288	0.0811	0.0741	0.0418		
Fasteners	0.1345	0.0918	0.0828	0.0333		
Furnishings	0.4039	0.0777	0.1067	0.0329		
Labels	0.1132	0.0673	0.1077	0.0190		
Machines	0.3286	0.2838	0.3333	0.3026		
Paper	0.1290	0.0646	0.1023	0.0316		
Phones	0.1220	0.1603	0.1000	0.2000		
Storage	0.1238	0.0711	0.0922	0.0308		
Supplies	0.1444	0.0679	0.1034	0.0377		
Tables	0.2625	0.3738	0.2225	0.2000		

Figure 11: Discounts by Sub-Category and Region

Therefore, the study shows how high discounts reduce profitability, and therefore the need for reasonable and strategic discounting to improve organizational performance.

Selection of Data Mining Algorithm and Data Pre-Processing

The K-Means algorithm is selected for further study as it finds natural grouping in data, such as customer segments, regional patterns, and product groups. Previous visualisations focused on variations in profitability, discounting patterns, and regional trends. For example, the general average discount in the Central/ East geographic regions tended to higher discount rate with the lower profit, and the Tables sub-index presented the tendency to lower added value. Clustering can also help identify the underlying trends in these variables to nurture desired strategies.

K-Means divides the dataset into clusters and minimises the within-cluster sum of squares (WCSS). It assigns data points to the nearest cluster centroid which is repeatedly updated and modified until the points stabilise. This process helps to discover the relations and clusters which are also valuable for decision making.

The dataset is ideal for clustering because of numerical variables like sales, profit, and discount, along with categorical variables like region and category. Graphs like scatter plots and heat maps (Figures 5, 10, 11) show that segmentation can enhance understanding of discounts, regional performance, and profitability connections.

Although It is a clean dataset, pre-processing enhances the result. Numerical variables like sales, profit, and discount should be normalised due to the sensitivity to the scales. Variables like region should be encoded into numerical format as they are categorical, and any outliers should be examined and capped if necessary. Also, some variables are not important for clustering so I dropped them. These steps make the flow of clustering possible and useful for solving problems such as underperforming sub-categories and regional inefficiencies. (Central = 0, East = 1, South = 2, West = 3)

```
normalise.py > ...
1  import pandas as pd
2  from sklearn.preprocessing import MinMaxScaler, LabelEncoder
3
4  file_path = "Sample - Superstore.csv"
5  raw_data = pd.read_csv(file_path, encoding='latin1')
6
7  columns_to_normalise = ['Sales', 'Profit', 'Discount']
8  scaler = MinMaxScaler()
9  raw_data[columns_to_normalise] = scaler.fit_transform(raw_data[columns_to_normalise])
10
11 columns_to_encode = ['Region', 'Category', 'Sub-Category']
12 label_encoder = LabelEncoder()
13 for col in columns_to_encode:
14     raw_data[col] = label_encoder.fit_transform(raw_data[col])
15
16 columns_to_exclude = ['Order ID', 'Customer ID', 'Order Date', 'Ship Date', 'Product Name', 'Customer Name', ]
17 raw_data = raw_data.drop(columns=[col for col in columns_to_exclude if col in raw_data.columns], errors='ignore')
18
19 output_file_path = 'Superstore_Preprocessed.csv'
20 raw_data.to_csv(output_file_path, index=False)
```

Figure 12: Screenshot of Data Pre-Processing Codes in Python

Data Mining

The first iteration of K-Means clustering used 15 variables in the dataset - except 6 columns that I removed in Figure 12 because it was obvious they did not have any effect in the analysis- with K = 3 to provide a baseline analysis. The algorithm achieved a within-cluster of squared errors (WCSS) of 35,754.93. Clusters are nearly balanced, with 39%, 29%, and 32% of data points in each group (Figure 13).

```
Number of iterations: 8
Within cluster sum of squared errors: 35745.93209166218

Initial starting points (random):

Cluster 0: 8826, 'First Class', Consumer, 'United States', Philadelphia, Pennsylvania, 19134, 1, OFF-LA-10003190, 1, 10, 0.000286, 3, 0.25, 0.440167
Cluster 1: 2743, 'Standard Class', Consumer, 'United States', 'San Diego', California, 92037, 3, FUR-FU-10004671, 0, 9, 0.001541, 2, 0, 0.440895
Cluster 2: 550, 'Second Class', Corporate, 'United States', Chicago, Illinois, 60653, 0, OFF-BI-10004738, 1, 3, 0.000482, 3, 1, 0.438863

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute          Full Data          Cluster#
                   (9994.0)          0
                   (3871.0)          1
                   (2943.0)          2
                   (3180.0)

=====
Row ID              4997.5             4968.5373           4969.7278           5058.4585
Ship Mode           Standard Class      Standard Class       Standard Class       Standard Class
Segment             Consumer            Consumer             Consumer             Corporate
Country             United States       United States         United States         United States
City                New York City       New York City         Los Angeles           Houston
State               California           New York              California             Texas
Postal Code          55190.3794          27265.0349           91164.2701           55890.9607
Region              1.5706             1.0827              2.9698              0.8698
Product ID          OFF-PA-10001970     TEC-AC-10003628      FUR-FU-10004270      OFF-BI-10000301
Category            0.9726             0.9799              0.964              0.9717
Sub-Category        7.5905             7.6079              7.6582              7.5066
Sales               0.0101             0.0108              0.0099              0.0095
Quantity            3.7896             3.7189              3.8192              3.8481
Discount            0.1953             0.1655              0.1239              0.2975
Profit              0.4419             0.4425              0.4423              0.4408

Time taken to build model (full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      3871 ( 39%)
1      2943 ( 29%)
2      3180 ( 32%)
```

Figure 13: Result of the first iteration

Key differences are discount and region. Cluster 2 has the highest average discount with 0.2975, while Cluster 1 has the lowest with 0.1239. Region separates clusters geographically, with cluster 0 showing central/east, cluster 1 for the west, and cluster 2 representing the south. Variables like profit, sales, and identifier have limited impact on grouping. The first iteration gives us a foundational understanding of broad patterns but tells us the need for refinement to better capture regional trends.

To do the second iteration, the number of K will be increased to 4 to align with four regions and provide more insight into regional performance. Also, those variables that have less impact like city, country, product ID, row ID, and postal code will be removed to enhance clustering precision.

As Figure 14 represents, this iteration achieved a within-cluster sum of squared errors (WCSS) of 16,127.05, an improvement from the first iteration, showing more compact and meaningful clusters.

```
kMeans
=====

Number of iterations: 20
Within cluster sum of squared errors: 16127.051948696593

Initial starting points (random):

Cluster 0: 'First Class',Consumer,Pennsylvania,1,1,10,0.000286,3,0.25,0.440167
Cluster 1: 'Standard Class',Consumer,California,3,0,9,0.001541,2,0,0.440895
Cluster 2: 'Second Class',Corporate,Illinois,0,1,3,0.000482,3,1,0.438863
Cluster 3: 'First Class',Consumer,Michigan,0,0,16,0.018171,2,0,0.444667

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute          Full Data          Cluster#
                   (9994.0)          0          1          2          3
                   (1622.0)        (3900.0)        (1722.0)        (2750.0)
=====
Ship Mode          Standard Class Standard Class Standard Class Second Class Standard Class
Segment            Consumer      Consumer      Consumer      Corporate      Consumer
State              California    Texas         California    Texas         New York
Region             1.5706       0.4044       2.7446       1.4355       0.6782
Category           0.9726       1.0025       0.9685       0.9936       0.9476
Sub-Category       7.5905       6.1868       7.9626       6.5058       8.5698
Sales              0.0101       0.0079       0.0104       0.0099       0.0113
Quantity           3.7896       3.7417       3.8469       3.8188       3.7182
Discount           0.1953       0.5516       0.1328       0.1887       0.0778
Profit             0.4419       0.4371       0.4422       0.4421       0.4442

Time taken to build model (full training data) : 0.04 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      1622 ( 16%)
1      3900 ( 39%)
2      1722 ( 17%)
3      2750 ( 28%)
```

Figure 14: Result of the second iteration

In this iteration values of the region indicate that it seems to separate clusters geographically with their differences, while the category and quantity values are the same among clusters. Also, sales and profit are nearly equal but I keep it because it is an important metric in my analysis. In the sub-category value, the difference between values is seen, so it is a good idea to keep it for the next iteration. Also, variables like ship mode and segment have nearly the same values across the cluster. The second iteration provided clear insight into regional and suc-category patterns.

For the final iteration, I removed those values mentioned earlier. The third iteration achieved a WCSS of 1,381.43, showing a significant reduction from the second iteration (Figure 15).

```

KMeans
=====

Number of iterations: 14
Within cluster sum of squared errors: 1381.4298799531266

Initial starting points (random):
Cluster 0: 1,10,0.000286,3,0.25,0.440167
Cluster 1: 3,9,0.001541,2,0,0.440895
Cluster 2: 0,3,0.000482,3,1,0.438863
Cluster 3: 0,16,0.018171,2,0,0.444667

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute      Full Data      Cluster#
                (9994.0)      0          1          2          3
=====
Region         1.5706         0.872       2.7575     0.8606     0.5879
Sub-Category   7.5905         2.3663     8.4901     3.9989     12.0567
Sales          0.0101         0.0118     0.0108     0.0042     0.0096
Quantity       3.7896         3.8107     3.8025     3.9177     3.7081
Discount       0.1953         0.1273     0.1175     0.8891     0.1472
Profit         0.4419         0.4442     0.4425     0.4344     0.4415

Time taken to build model (full training data) : 0.03 seconds

=== Model and evaluation on training set ===

Clustered Instances
0      2367 ( 24%)
1      4107 ( 41%)
2       875 (  9%)
3      2645 ( 26%)

```

Figure 15: Result of the Third Iteration

Cluster 0 with a region value of 0.872 shows an overlap between central and east regions and a sub-category value of 2.36 shows us the most important sub-category is art (2 maps to art). The sales are highest among all the groups with 0.0118, while profit is at 0.4442. Discount at 0.1237 demonstrates moderate discounting. This cluster shows balanced sales and profit strategies in central and east regions with controlled discounting.

With a region value of 2.7575, cluster 1 aligns closely with the west region. In this cluster, Furnishing and fasteners are the most important sub-categories with a value of 8.49. Sales and profit are 0.0108 and 0.4424, while it has the lowest discount rate across clusters (0.1175), showing high margin sales. The results reflect that in the west region, furnishing and fasteners are sold with minimal discount leading to profitability.

Cluster 2 with a region value of 0.8606 is more likely to point to the east region. The centroid value of 3.99 for the sub-category shows that the bookcases (4 maps to bookcases) are the most important in this cluster. This cluster has the lowest sales with a value of 0.0042, while profit is normal with a value of 0.4344. With an average discount of 0.8891, it has the highest cluster average. This shows that the aggressive discounting policy did not significantly affect sales.

A value for a region of 0.5879, indicates an overlap between east and central regions. This cluster is dominated by papers with a sub-category value of 12.06. A discount rate of 0.1472, balancing sales and profitability with values of 0.0096 and 0.4415. These values suggest that papers with moderate discounts led to balanced sales and profit in the east and central regions.

The final iteration gives us a clarified vision by focusing on sub-categories and regions, with the lowest value of WCSS (1,381) among iterations. Each cluster shows unique sub-categories and regional patterns, mentioning various policies: moderate discounts and balanced sales in cluster 0, minimal discount and high margin sales in cluster 1, high discount policy without remarkable sales in cluster 2, and balanced discounts led to stable sales and profit in cluster 3. These results highlight the diverse regional strategies required for optimising profitability.

Data Ethics

Analyzing and using business data therefore raises serious ethical, legal, and professional issues. From an ethical standpoint, data privacy and confidentiality must be considered most importantly. Information privacy for customers requires organizations to store customer information securely, gain permission to use the information and refrain from using it in improper ways. Inconclusiveness in data analysis is a betrayal of trust and a clear ethical concern.

From a legal perspective, it is now obligatory to abide by the data protection rules of the GDPR and the UK Data Protection Act. These laws enshrine duties for the protection of the information, proper usage of the information, and gaining prior permission for the collection and processing. Sanctions include major fines as well as legal consequences.

Ethically, analysts should provide professionalism in their work through out-shadowing bias and maintaining the overall, unprejudiced results. Sloppy meanings or malicious interpretations of data undermine confidence and diminish professional standing. There is no doubt that transparency of methods and decision-making mechanisms is crucial for the accountability of institutions.

Referring to the lecture on data ethics, the FAT principles must be followed: fairness, accountability, and transparency. For example, it is important to constantly check the algorithms such as K-Means clustering in order to avoid scalability, where some regions or sub-categories are skewed on one side unduly. Achieving these goals helps to regulate the use of data and to protect stakeholders, which contributes to trusting data analysis results.

Conclusion

The graphs provide a complete overview of regional performance, sub-category trends, and the effect of discounts on profit. The most important findings are high sales in the central region but with low profitability due to the high discounts and underperforming categories like tables and supplies. This analysis shows that sales are at the top with a moderate discount level but decrease sharply beyond 30% discounts. Also, sub-categories such as copiers and phones have a good profit with controlled discounts, while others like tables, performed poorly despite heavy discounting. These insights mention the need for regional and product focused strategies.

The K-Means clustering ran three times and gave us valuable segmentation insights. The final iteration shows the best fit to data with four clusters and the lowest WCSS value (1,381.43). This analysis tells about unique patterns, in the central and east regions sales and profit are balanced in cluster 0. Cluster 1 shows high margin sales in the west region, cluster 2 found high discounting without any bold impact on sales in the east region, and cluster 3 shows balanced profitability for papers in overlapping central and east regions. This clustering gave us a clearer view of regional and sub-category performance and enabled actionable recommendations.

This analysis and its result present useful insight to managers and investors in the business. Regional trends can help pricing and discount strategy, such as keeping discount at 20% in the highest case for those sub-categories have low profit and promoting high-margin items such as copiers. The marketing team can use sub-category insights to focus on profitable products, and the operation team can work on underperforming sub-category to reduce its cost. For example, using a dynamic pricing approach for high cost items in the central region can increase profit margins. Also, using promotions with seasonal trends could smooth profit across quarters. These results give us a foundation for informed decision making and enable the company to optimise profitability and maintain competitive advantage.